

CHAPTER 2

2.1 Requirement Analysis

Requirement analysis, also known as requirement engineering, is the process of identifying, analyzing, and documenting the needs and expectations of users for a new or modified system. These requirements must be specific, measurable, and clearly defined to ensure the system meets user expectations.

In the case of the *Web-Based Timetable and Classroom Management System*, requirement analysis focuses on understanding how administrators, teachers, and students interact with the system to efficiently manage class schedules, classroom allocations, and related academic activities. This process ensures that the developed system enhances productivity, reduces scheduling conflicts, and provides an organized academic environment.

2.2 Functional Requirements

2.2.1 Admin Requirements

- **Login:** The Admin will log in using a valid username and password to access the system.
- **Manage Teachers and Students:** The Admin can add, edit, or delete teacher and student accounts.
- **Manage Classrooms:** The Admin can add, modify, or delete classroom information including room numbers and capacities.
- **Generate Timetable:** The Admin can automatically generate or manually create timetables for different classes and sections.
- **Resolve Scheduling Conflicts:** The system will alert the Admin of overlapping classes or unavailable rooms.
- **View and Print Timetable:** The Admin can view and print final timetables for departments, teachers, and students.
- **Logout:** The Admin can securely log out of the system to end the session.

2.2.2 Teacher Requirements

- Login: Teachers can log in to their accounts using valid credentials.
- View Assigned Timetable: Teachers can view their timetable showing classes, subjects, and room details.
- Request Timetable Change: Teachers can submit requests to the Admin for timetable adjustments (e.g., class timing or room changes).
- View Classroom Details: Teachers can view information about classrooms such as room number, location, and capacity.
- Check Schedule Conflicts: Teachers can check for overlapping sessions or schedule conflicts.
- Logout: Teachers can securely log out of the system after use.

2.2.3 Student Requirements

- Login: Students can log in to their portal using valid credentials.
- View Personal Timetable: Students can view their class timetable including course names, teachers, and classrooms.
- View Classroom Allocation: Students can check the room number and details for each class session.
- Print Timetable: Students can print or download their class timetable for convenience.
- Logout: Students can securely log out after checking their schedule.

2.3 Non-Functional Requirements

Here's your non-functional requirements written neatly in a table format:

Non-Functional Requirement	Description
Security	The system should use secure authentication methods and data encryption to protect user information.
Performance	The system should provide quick access to data and generate timetables

	efficiently.
Reliability	The system should operate accurately without data loss or duplication.
Usability	The interface should be simple, intuitive, and user-friendly for all users.
Scalability	The system should be capable of handling increased data as the institution grows.
Availability	The system should be available 24/7 with minimal downtime.
Maintainability	The system should be easy to update and maintain by administrators.

2.4 Use Case Fully Dressed Style

The Web-Based Timetable and Classroom Management System includes the following major use cases:

- Login
- Manage Teachers and Students
- Manage Classrooms
- Generate Timetable
- View Timetable
- Request Timetable Change
- Logout

Table 2.1: Use Case – Login

Use Case Name	Login
Level	User-goal
Scope	Timetable and Classroom Management System
Primary Actor	Admin, Teacher, Student
Stakeholders and Interests	All users: Successful login to access respective modules. System: Authenticate users securely.
Pre-condition	User must be registered and have valid credentials.
Post-condition	User successfully logs in and accesses their dashboard.

Main Success Scenario	1. User navigates to the login page. 2. Enters valid credentials. 3. System verifies details. 4. On success, user is directed to their respective dashboard. 5. System ensures secure session management.
Frequency of Occurrence	Multiple times daily.
Technology	• HTML • CSS • JavaScript (Frontend) • MySQL (Database)
Special Requirements	Passwords should be encrypted; invalid login attempts should be limited.

Table 2.2: Use Case – Manage Teachers and Students

Use Case Name	Manage Teachers and Students
Level	User-goal
Scope	Timetable and Classroom Management System
Primary Actor	Admin
Stakeholders and Interests	Admin: Manage teacher and student profiles efficiently. System: Ensure records are updated in the database.
Pre-condition	Admin must be logged in.

Post-condition	Updated user data saved successfully in the database.
Main Success Scenario	1. Admin navigates to “Manage Users.” 2. Adds, edits, or deletes a user. 3. System validates inputs. 4. Records updated successfully.
Frequency of Occurrence	As needed by Admin.
Technology	• HTML • MySQL
Special Requirements	Prevent duplicate entries and ensure data integrity.

Table 2.3: Use Case – Generate Timetable

Use Case Name	Generate Timetable
Level	User-goal
Scope	Timetable and Classroom Management System
Primary Actor	Admin
Stakeholders and Interests	Admin: Efficiently create schedules for teachers and students. System: Avoid conflicts in class timing and room availability.
Pre-condition	Courses, teachers, and classrooms must be registered.
Post-condition	Timetable is successfully generated and saved.

Main Success Scenario	1. Admin selects “Generate Timetable.” 2. System checks all data. 3. Timetable is generated automatically. 4. Conflicts are highlighted if any. 5. Final timetable saved and available for all users.
Frequency of Occurrence	Once per semester or as needed.
Technology	MySQL
Special Requirements	Avoid overlapping classes; allow manual editing after generation.

Table 2.4: Use Case – View Timetable

Use Case Name	View Timetable
Level	User-goal
Scope	Timetable and Classroom Management System
Primary Actor	Teacher, Student
Stakeholders and Interests	Teachers and students need real-time access to their timetables. System: Display accurate and updated information.
Pre-condition	Timetable must be generated and stored in the system.
Post-condition	User successfully views their timetable.
Main Success Scenario	1. User logs in. 2. Navigates to “View Timetable.” 3. System retrieves and displays the relevant timetable.
Frequency of Occurrence	Daily
Technology	• HTML

	• CSS • MySQL
Special Requirements	Timetable should auto-refresh when updates are made by Admin.

Table 2.5: Use Case – Request Timetable Change

Use Case Name	Request Timetable Change
Level	User-goal
Scope	Timetable and Classroom Management System
Primary Actor	Teacher
Stakeholders and Interests	Teacher: Request schedule adjustments Admin: Review and approve/reject requests. System: Track changes and update timetables.
Pre-condition	Timetable must be generated and teacher logged in.
Post-condition	Request stored and sent to Admin for action.
Main Success Scenario	1. Teacher selects “Request Change.” 2. Enters reason and desired time/room. 3. System sends request to Admin. 4. Admin reviews and approves/rejects it. 5. Timetable updates accordingly.
Frequency of Occurrence	Occasionally
Technology	HTML

Special Requirements	Track all change requests for audit purposes.
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Table 2.6: Use Case – Logout

Use Case Name	Logout
Level	User-goal
Scope	Timetable and Classroom Management System
Primary Actor	• Admin • Teacher • Student
Stakeholders and Interests	All users: Securely end session and protect data. System: Ensure proper session termination.
Pre-condition	User must be logged in.
Post-condition	User session terminated; redirected to login page.
Main Success Scenario	1. User clicks “Logout.” 2. System ends session. 3. Redirects user to login page.
Frequency of Occurrence	Multiple times daily

Technology	• HTML • CSS
Special Requirements	Automatic logout after inactivity period.